SOEN 342: Software Requirements and Deployment
Project - Iteration I
System Modelling and Requirements Specification
Fall 2026

Date posted: Tuesday, October 6th, 2026.
Date due: Friday, October 23rd, 2026, by 23:59.
Weight: 20% of the overall grade (in total).

Deliverables (Artifacts)

1. Use-case diagram, including critical and non-critical use-cases.

Created using https://www.plantuml.com/plantuml/uml/

```
@startuml
title Appointment Tracking System -
Use-Case Diagram

left to right direction
skinparam shadowing false
skinparam backgroundColor white
skinparam defaultFontName Arial
skinparam packageStyle rectangle
skinparam linetype polyline
skinparam nodesep 30
skinparam ranksep 35

actor Patient as P
actor "Authorized Clinic Staff" as Staff
actor "Administrative Staff" as Admin
actor "Healthcare Professional" as HP
actor "Family Physician" as FP
actor "Specialist Physician" as SP
actor Nurse as N
actor "Other Healthcare Professional" as O

Staff <|-- Admin
Staff <|-- HP
HP <|-- FP
HP <|-- SP
HP <|-- N
HP <|-- O

rectangle "Appointment Tracking System" {
  usecase "Request Appointment" as Request
  usecase "Book Appointment" as Book
  usecase "Confirm Appointment" as
Confirm
  usecase "Reschedule Appointment" as
Reschedule
  usecase "Cancel Appointment" as Cancel
  usecase "Register Patient Arrival" as
Arrival
  usecase "Register / Update Patient" as
PatientInfo
  usecase "Search Patient" as Search
  usecase "Maintain Professional,
Clinic,\nSpecialty and Service Information"
as Maintain
  usecase "Manage Professional Schedule"
as Schedule
  usecase "Conduct Consultation" as Consult
```

```
  usecase "Record Clinical Information" as
Clinical
  usecase "Review Clinical Records" as
Review
  usecase "Issue Prescription" as Rx
  usecase "Issue Referral" as Referral
  usecase "Request Follow-up Appointment"
as Follow
  usecase "Record Service Fee" as Fee
  usecase "Generate Invoice" as Invoice
  usecase "Register Payment" as Payment
}

P -- Request
Staff -- Request
Staff -- Book
Admin -- Confirm
Admin -- Reschedule
Admin -- Cancel
Admin -- Arrival
Admin -- PatientInfo
Admin -- Search
Staff -- Maintain

HP -- Schedule
HP -- Consult
HP -- Review
HP -- Rx
HP -- Referral
HP -- Follow

Admin -- Fee
Admin -- Invoice
Admin -- Payment

Consult ..> Clinical : <<include>>
Rx ..> Consult : <<extend>>\n[medication
needed]
Follow ..> Consult :
<<extend>>\n[follow-up needed]

note right of Book
  CRITICAL UC1
  Check availability, scheduling
  constraints and referral rules.
end note

note right of Consult
  CRITICAL UC2
end note

note right of Referral
  CRITICAL UC3
end note

note bottom of Request
  Patients may request appointments.
  Direct patient CLI access is not specified;
  requests may be mediated by clinic staff.
end note

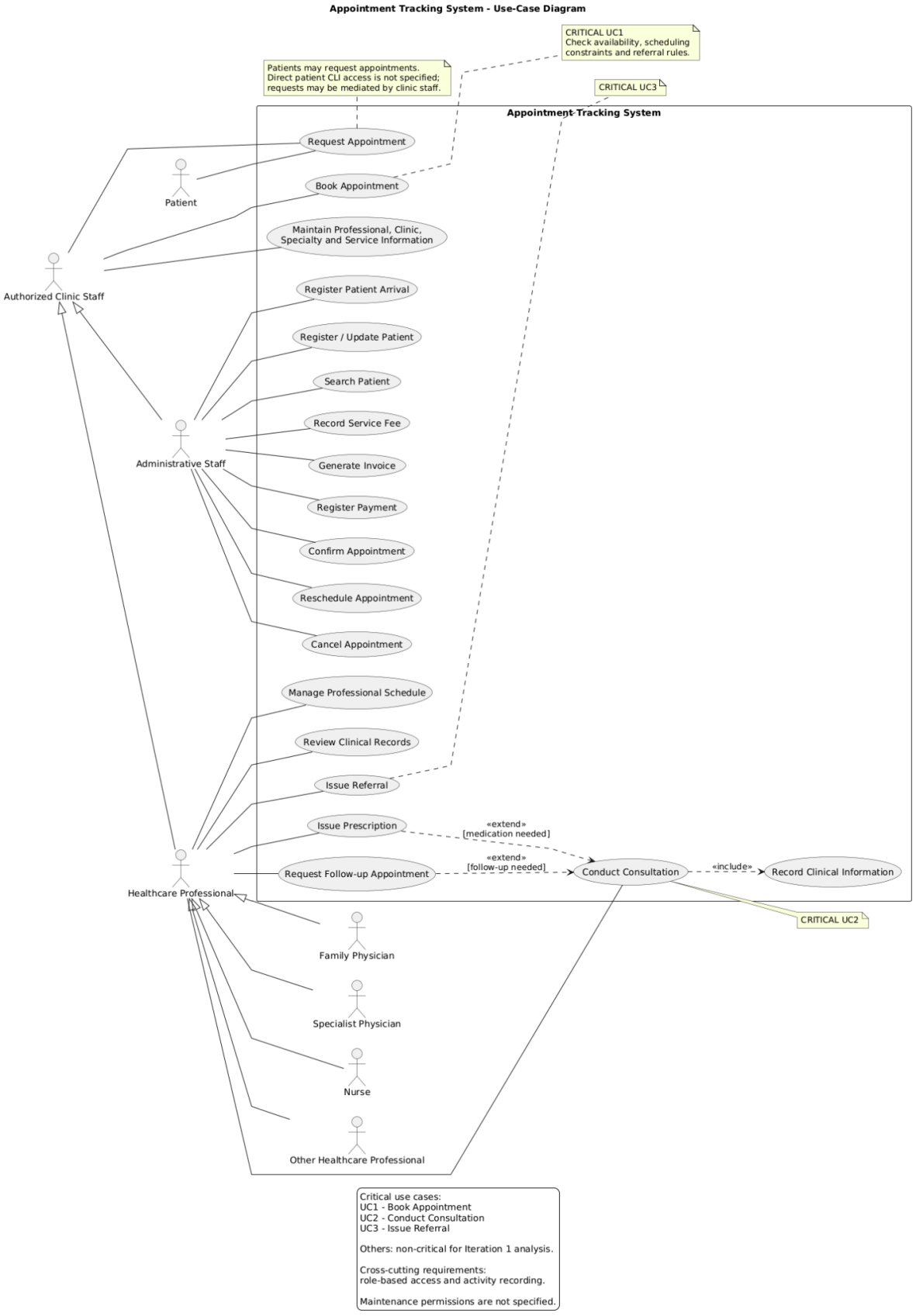
legend bottom
  Critical use cases:
  UC1 - Book Appointment
  UC2 - Conduct Consultation
  UC3 - Issue Referral

  Others: non-critical for Iteration 1 analysis.

  Cross-cutting requirements:
  role-based access and activity recording.

  Maintenance permissions are not specified.
endlegend

@enduml
```


Appointment Tracking System - Use-Case Diagram
CRITICAL UC1
Check availability, scheduling constraints and referral rules.
Patients may request appointments.
Direct patient CLI access is not specified;
requests may be mediated by clinic staff.
CRITICAL UC3
Appointment Tracking System
Request Appointment
Book Appointment
Maintain Professional, Clinic, Specialty and Service Information
Register Patient Arrival
Register / Update Patient
Search Patient
Record Service Fee
Generate Invoice
Register Payment
Confirm Appointment
Reschedule Appointment
Cancel Appointment
Manage Professional Schedule
Review Clinical Records
Issue Referral
Issue Prescription
Request Follow-up Appointment
«extend»
[medication needed]
«extend»
[follow-up needed]
Conduct Consultation
«include»
Record Clinical Information
CRITICAL UC2
Patient
Authorized Clinic Staff
Administrative Staff
Healthcare Professional
Family Physician
Specialist Physician
Nurse
Other Healthcare Professional
Critical use cases:
UC1 - Book Appointment
UC2 - Conduct Consultation
UC3 - Issue Referral
Others: non-critical for Iteration 1 analysis.
Cross-cutting requirements:
role-based access and activity recording.
Maintenance permissions are not specified.

2. UML domain model

Created using https://www.plantuml.com/plantuml/uml/

```
@startuml
title Appointment Tracking System - UML Domain Model

skinparam shadowing false
skinparam backgroundColor white
skinparam defaultFontName Arial
skinparam classAttributeIconSize 0
skinparam linetype polyline
skinparam nodesep 45
skinparam ranksep 65

hide methods
hide circle

' ============================
' PATIENT INFORMATION
' ============================

class Patient {
  patientID : String
  firstName : String
  lastName : String
  dateOfBirth : Date
  phoneNumber : String
}

class InsuranceCoverage {
  insuranceNumber : String
  type : String
}

' ============================
' HEALTHCARE PROFESSIONALS
' ============================

abstract class HealthcareProfessional {
  professionalID : String
  name : String
  professionalInformation : String
}

class FamilyPhysician
class SpecialistPhysician
class Nurse
class OtherHealthcareProfessional

class MedicalSpecialty {
  name : String
}

class ClinicLocation {
  locationID : String
  address : String
}

class MedicalService {
  serviceID : String
  name : String
  requiresReferral : Boolean
}

class ProfessionalSchedule {
  scheduleID : String
  startDateTime : DateTime
  endDateTime : DateTime
}

' ============================
' APPOINTMENTS
' ============================
```

```
class Appointment {
  appointmentID : String
  dateTime : DateTime
  status : String
}

class PatientArrival {
  arrivalDateTime : DateTime
}

' ============================
' CONSULTATIONS
' ============================

class Consultation {
  consultationID : String
  dateTime : DateTime
}

class ClinicalRecord {
  recordID : String
  observations : String
  diagnoses : String
  treatmentRecommendations : String
  followUpRequirements : String
}

' ============================
' REFERRALS
' ============================

class Referral {
  referralID : String
  clinicalReason : String
  priority : String
  status : String
}

' ============================
' PRESCRIPTIONS
' ============================

class Prescription {
  prescriptionID : String
  issueDate : Date
}

class PrescriptionItem {
  dosageInstructions : String
}

class Medication {
  medicationID : String
  name : String
}

' ============================
' BILLING
' ============================

class ServiceFee {
  amount : Money
}

class Invoice {
  invoiceID : String
  issueDate : Date
  amount : Money
}

class Payment {
  paymentID : String
  date : Date
  amount : Money
}

' ============================
' ADMINISTRATIVE INFORMATION
' ============================
```

```
class AdministrativeStaff {
  staffID : String
  name : String
}

class ActivityRecord {
  activityID : String
  dateTime : DateTime
  description : String
}

' ==============================
' GENERALIZATION
' ==============================

HealthcareProfessional <|-- FamilyPhysician
HealthcareProfessional <|--
SpecialistPhysician
HealthcareProfessional <|-- Nurse
HealthcareProfessional <|--
OtherHealthcareProfessional

' ==============================
' PATIENT ASSOCIATIONS
' ==============================

Patient "1" -- "0..*" InsuranceCoverage :
Has

Patient "1" -- "0..*" Appointment : Has

Patient "1" -- "0..*" ClinicalRecord : Has

Patient "1" -- "0..*" Referral : Is-subject-of

' ==============================
' PROFESSIONAL ASSOCIATIONS
' ==============================

HealthcareProfessional "0..*" -- "0..*"
ClinicLocation : Practises-at

HealthcareProfessional "0..*" -- "0..*"
MedicalSpecialty : Has-specialty

HealthcareProfessional "1" -- "0..*"
ProfessionalSchedule : Has

ClinicLocation "1" -- "0..*"
ProfessionalSchedule : Applies-at

ClinicLocation "0..*" -- "0..*"
MedicalService : Offers

' ==============================
' APPOINTMENT ASSOCIATIONS
' ==============================

HealthcareProfessional "1" -- "0..*"
Appointment : Is-assigned-to

ClinicLocation "1" -- "0..*" Appointment :
Hosts

MedicalService "1" -- "0..*" Appointment :
Is-requested-in

Appointment "1" -- "0..1" PatientArrival :
Has

AdministrativeStaff "0..1" -- "0..*"
PatientArrival : Registers

Appointment "1" -- "0..1" Consultation :
Results-in

HealthcareProfessional "1" -- "0..*"
Consultation : Conducts
```

```
' ==============================
' CONSULTATION ASSOCIATIONS
' ==============================

Consultation "1" -- "0..*" ClinicalRecord : Documents

Consultation "1" -- "0..*" Prescription : Produces

' ==============================
' REFERRAL ASSOCIATIONS
' ==============================

HealthcareProfessional "1" -- "0..*" Referral : Issues

MedicalSpecialty "0..1" -- "0..*" Referral : Requests-specialty

HealthcareProfessional "0..1" -- "0..*" Referral : Requests-professional

Referral "0..1" -- "0..*" Appointment : Supports

' ==============================
' PRESCRIPTION ASSOCIATIONS
' ==============================

Prescription "1" -- "1..*" PrescriptionItem : Contains

Medication "1" -- "0..*" PrescriptionItem : Identifies

' ==============================
' BILLING ASSOCIATIONS
' ==============================

Patient "1" -- "0..*" Invoice : Receives

MedicalService "1" -- "0..*" ServiceFee : Has-applicable-fee

Invoice "0..*" -- "1..*" ServiceFee : Includes

Invoice "1" -- "0..*" Payment : Receives

' ==============================
' ACTIVITY RECORDING
' ==============================

AdministrativeStaff "0..1" -- "0..*" ActivityRecord : Performs

HealthcareProfessional "0..1" -- "0..*" ActivityRecord : Performs

' ==============================
' CONSTRAINTS
' ==============================

note right of Patient
  patientID is unique.
  Insurance coverage is optional.
end note

note right of HealthcareProfessional
  Referring professionals may be
  external and may not practise
  at a clinic location.
end note

note right of MedicalService
  A service may require a referral.
  Uninsured services may have
  applicable fees.
end note
```

```
note right of Appointment
  Booking must satisfy professional
  availability and scheduling rules.

  A valid referral is required
  when the service requires one.
end note

note right of Referral
  A referral identifies the referring
  professional, clinical reason,
  priority, and status.

  At least one requested specialty
  or professional is specified.

  A referral may result in
  multiple appointments.
end note

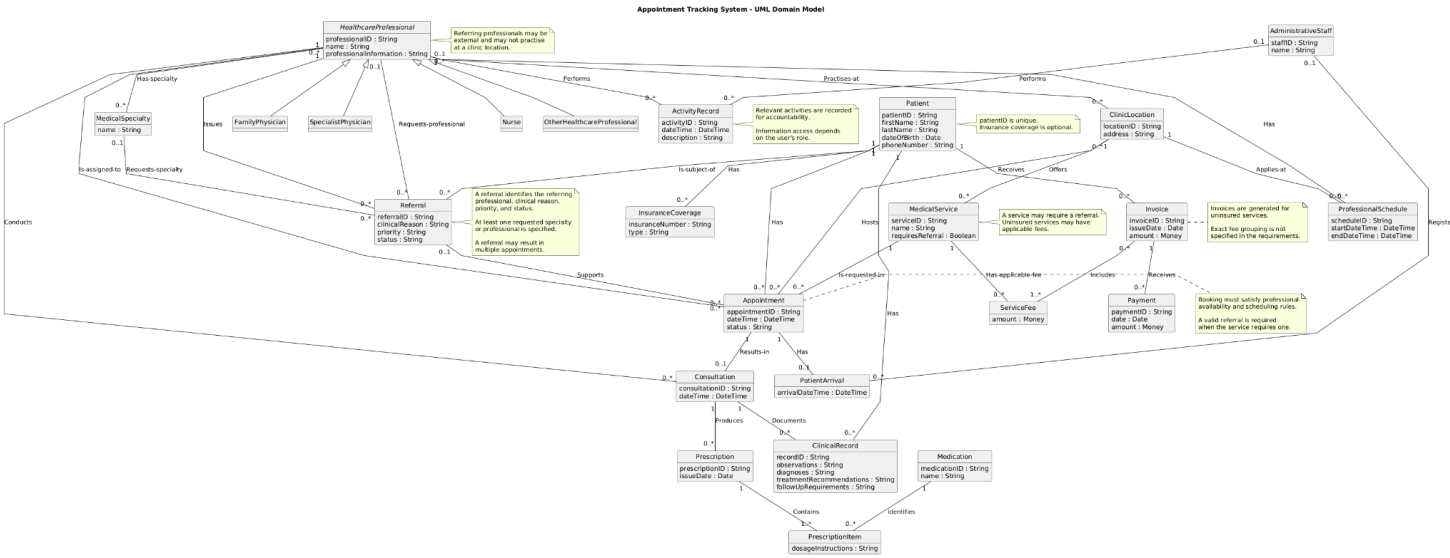
note right of Invoice
  Invoices are generated for
  uninsured services.

  Exact fee grouping is not
  specified in the requirements.
end note

note right of ActivityRecord
  Relevant activities are recorded
  for accountability.

  Information access depends
  on the user's role.
end note

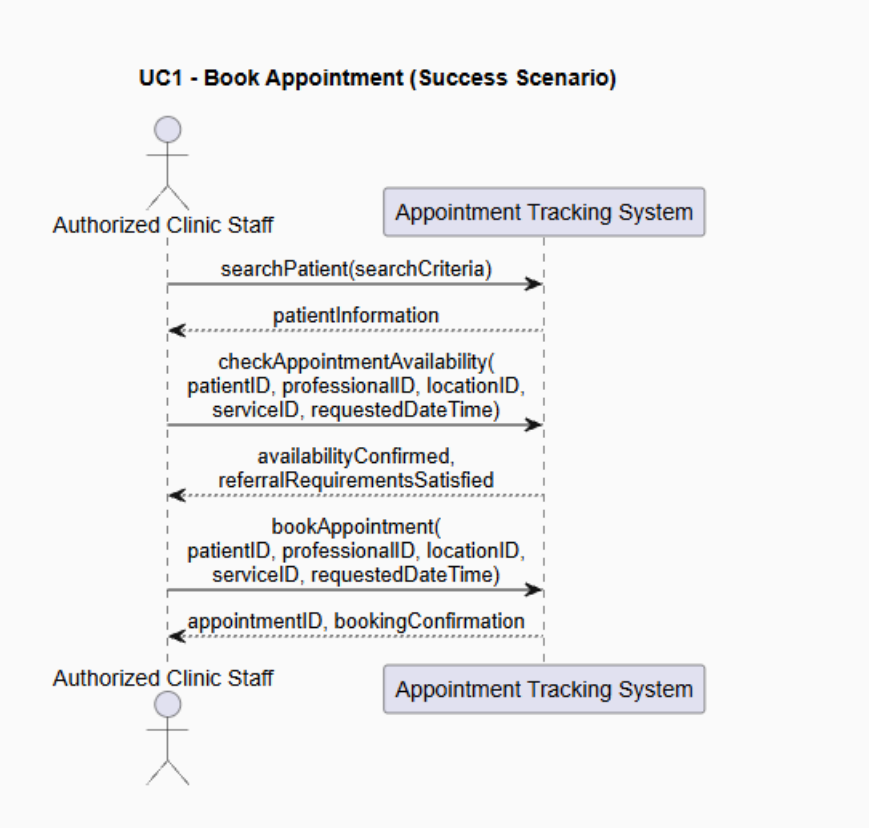
@enduml
```



3. System sequence diagram(s) to capture success and failure scenarios for critical use-cases.

4. Three fully-dressed scenarios for three critical use-cases, with at least one success and one failure case for each (an equivalent system sequence diagram to be provided in the above).
5. Identification of system operations and operation contracts.

3.System Sequence Diagram with both success and failure casses




```
@startuml
title UC1 - Book Appointment (Success
Scenario)

skinparam shadowing false
skinparam backgroundColor white
skinparam defaultFontName Arial
skinparam sequenceMessageAlign center

actor "Authorized Clinic Staff" as Staff
participant "Appointment Tracking System"
as System

Staff -> System:
searchPatient(searchCriteria)

System --> Staff: patientInformation

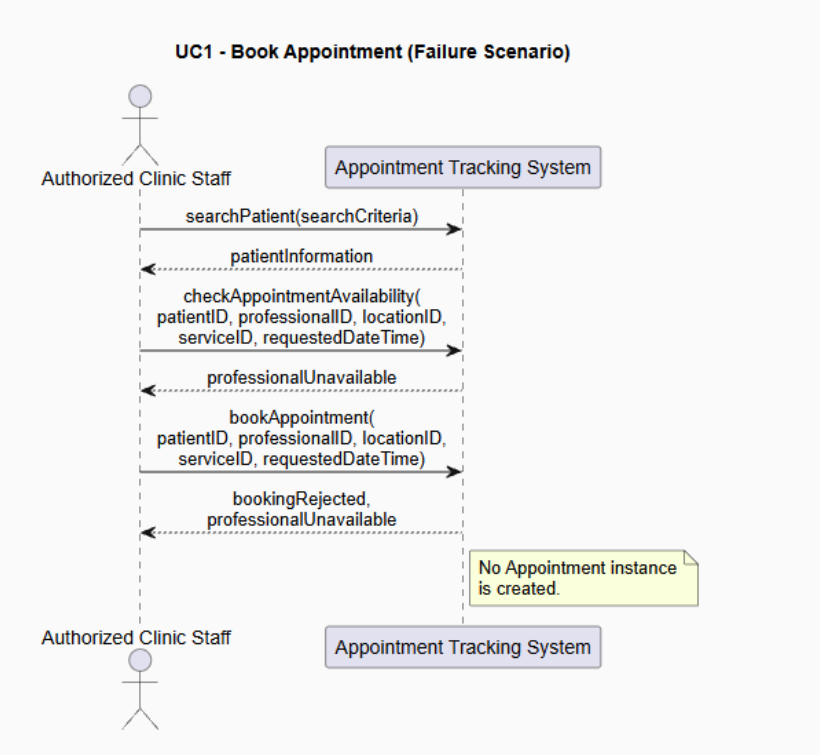
Staff -> System:
checkAppointmentAvailability(\npatientID,
professionalID, locationID,\nserviceID,
requestedDateTime)

System --> Staff:
availabilityConfirmed,\nreferralRequirement
sSatisfied

Staff -> System:
bookAppointment(\npatientID,
professionalID, locationID,\nserviceID,
requestedDateTime)
```

System --> Staff: appointmentID, bookingConfirmation

@enduml



@startuml
title UC1 - Book Appointment (Failure Scenario)

skinparam shadowing false
skinparam backgroundColor white
skinparam defaultFontName Arial
skinparam sequenceMessageAlign center

actor "Authorized Clinic Staff" as Staff
participant "Appointment Tracking System" as System

Staff -> System: searchPatient(searchCriteria)
System --> Staff: patientInformation

Staff -> System: checkAppointmentAvailability(\npatientID, professionalID, locationID,\nserviceID, requestedDateTime)

System --> Staff: professionalUnavailable

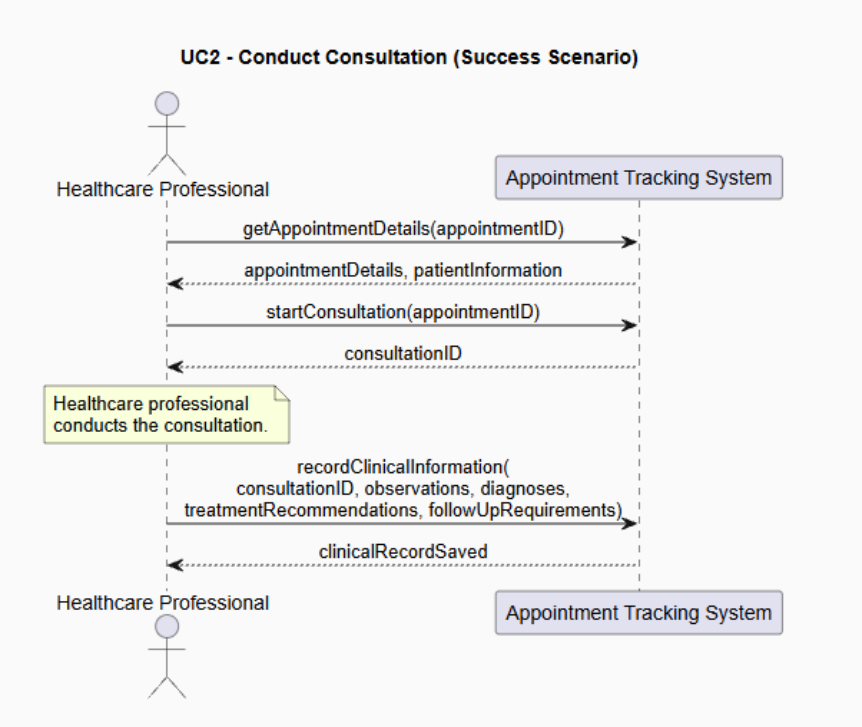
Staff -> System: bookAppointment(\npatientID, professionalID, locationID,\nserviceID, requestedDateTime)

System --> Staff: bookingRejected,\nprofessionalUnavailable

note right of System
  No Appointment instance
  is created.

end note

@enduml



@startuml
title UC2 - Conduct Consultation (Success Scenario)

skinparam shadowing false
skinparam backgroundColor white
skinparam defaultFontName Arial
skinparam sequenceMessageAlign center

actor "Healthcare Professional" as HP
participant "Appointment Tracking System" as System

HP -> System: getAppointmentDetails(appointmentID)
System --> HP: appointmentDetails, patientInformation

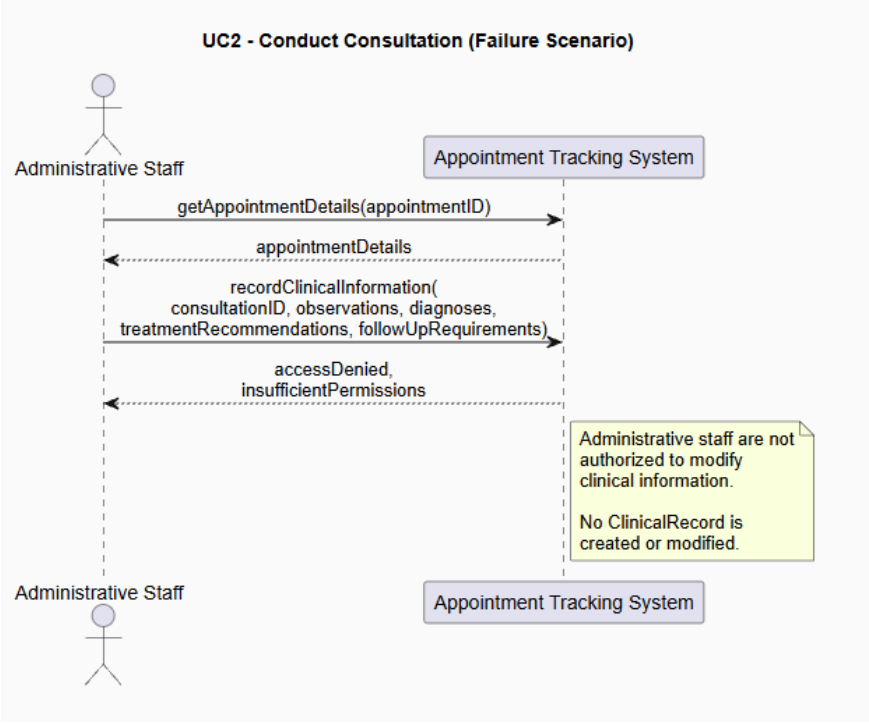
HP -> System: startConsultation(appointmentID)
System --> HP: consultationID

note over HP
  Healthcare professional
  conducts the consultation.
end note

HP -> System: recordClinicalInformation(\nconsultationID, observations, diagnoses,\ntreatmentRecommendations, followUpRequirements)

System --> HP: clinicalRecordSaved

@enduml




```
@startuml
title UC2 - Conduct Consultation (Failure Scenario)

skinparam shadowing false
skinparam backgroundColor white
skinparam defaultFontName Arial
skinparam sequenceMessageAlign center

actor "Administrative Staff" as Admin
participant "Appointment Tracking System" as System

Admin -> System: getAppointmentDetails(appointmentID)
System --> Admin: appointmentDetails

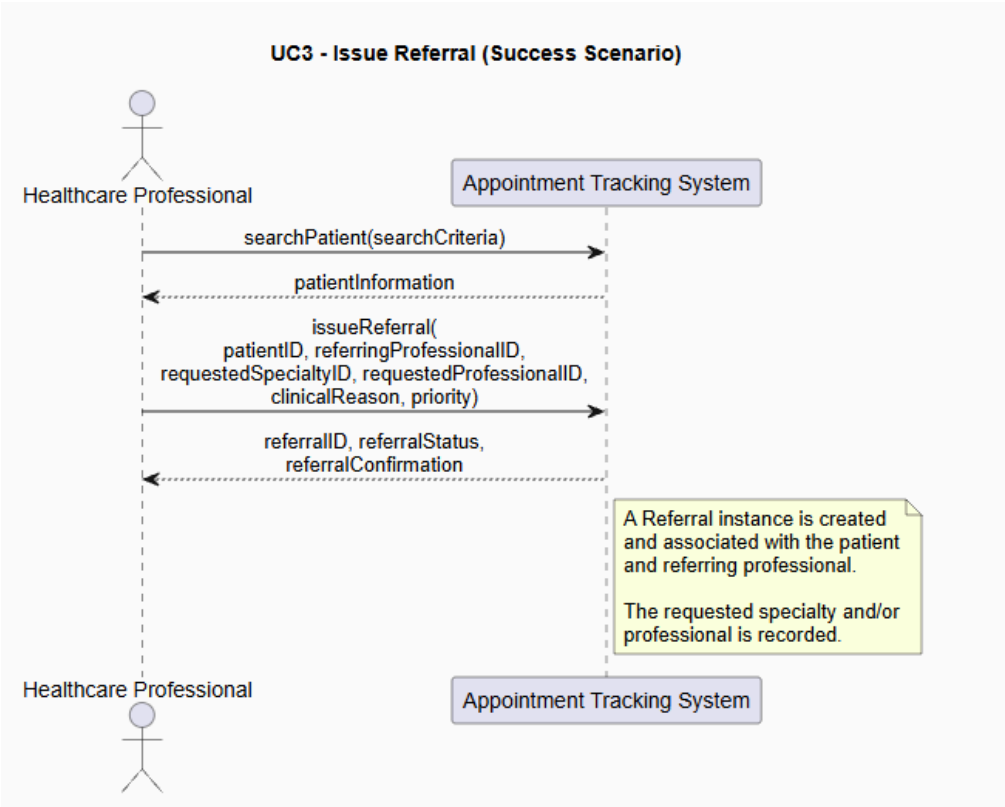
Admin -> System: recordClinicalInformation(\nconsultationID, observations, diagnoses,\ntreatmentRecommendations, followUpRequirements)

System --> Admin: accessDenied,\ninsufficientPermissions

note right of System
  Administrative staff are not
  authorized to modify
  clinical information.

  No ClinicalRecord is
  created or modified.
end note

@enduml
```




```
@startuml
title UC3 - Issue Referral (Success Scenario)

skinparam shadowing false
skinparam backgroundColor white
skinparam defaultFontName Arial
skinparam sequenceMessageAlign center

actor "Healthcare Professional" as HP
participant "Appointment Tracking System" as System

HP -> System: searchPatient(searchCriteria)
System --> HP: patientInformation

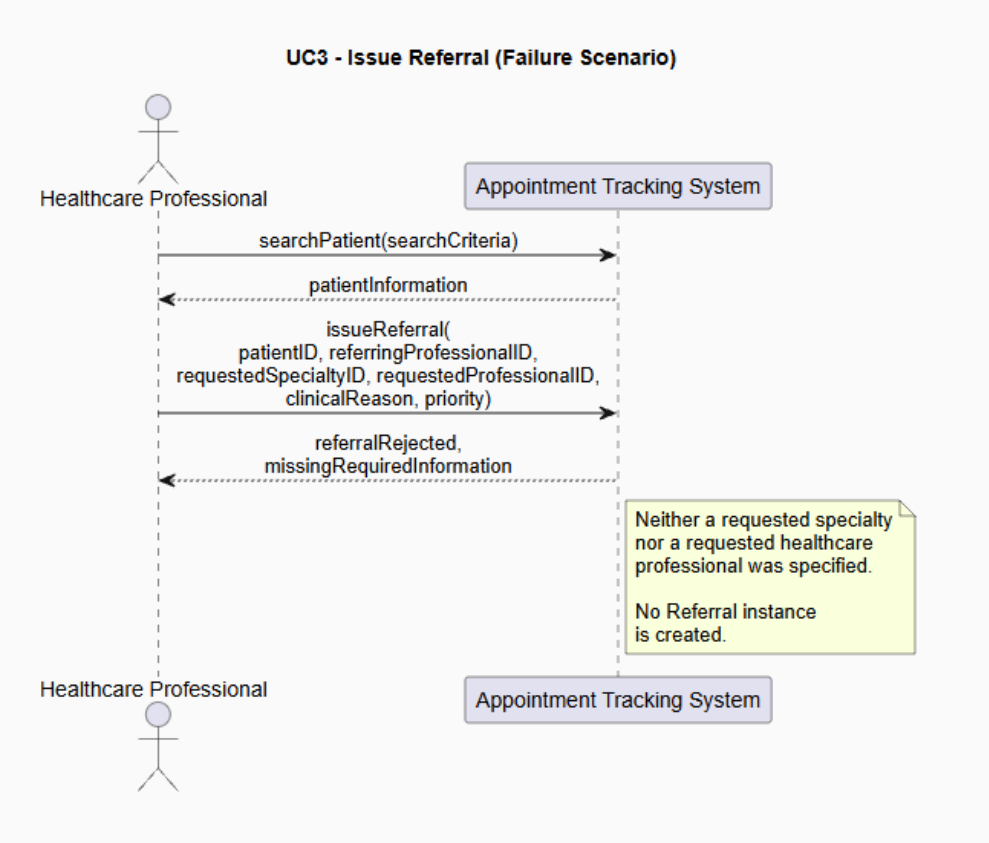
HP -> System: issueReferral(\npatientID, referringProfessionalID,\nrequestedSpecialtyID,
requestedProfessionalID,\nclinicalReason, priority)

System --> HP: referralID, referralStatus,\nreferralConfirmation

note right of System
  A Referral instance is created
  and associated with the patient
  and referring professional.

  The requested specialty and/or
  professional is recorded.
end note

@enduml
```




```
@startuml
title UC3 - Issue Referral (Failure Scenario)

skinparam shadowing false
skinparam backgroundColor white
skinparam defaultFontName Arial
skinparam sequenceMessageAlign center

actor "Healthcare Professional" as HP
participant "Appointment Tracking System"
as System

HP -> System:
searchPatient(searchCriteria)
System --> HP: patientInformation

HP -> System: issueReferral(\npatientID,
referringProfessionalID,\nrequestedSpecialt
yID,
requestedProfessionalID,\nclinicalReason,
priority)

System --> HP:
referralRejected,\nmissingRequiredInformati
on

note right of System
  Neither a requested specialty
  nor a requested healthcare
  professional was specified.

  No Referral instance
  is created.
end note

@enduml
```

4. Three fully-dressed scenarios for three critical use-cases, with at least one success and one failure case for each

Use Case 1 (Success):

Primary actor:
Clinic Staff
Stakeholders and Interest:
Patient:
Wants minimal wait time, wants a secure system to protect confidential info
Clinic Staff:
Wants fast patient lookup, fast checks for doctor availability, schedule constraints
And communication with receiving referrals between different clinics
Clinic Doctors:
Accurate non-complicated software to input days off and prevent bookings on those days.
Quick referral sending between clinics.

Preconditions:
- Clinic Staff or Clinic Doctors: are securely logged in with editing access to patient information or scheduling times.
- Clinic Doctors: want their schedules to be easily added to the system already.

Success Guarantee (Postconditions):
- An appointment is created and confirmed with a date, time, doctor and room.
- The patient is texted a link to the confirmed appointment with all the info clearly displayed.
- The updated system schedule now blocks off the confirmed time for the duration of the booked appointment.
- If referred the referral is connected to the booked appointment.

Main Success Scenario:
1. Patient requests an appointment at the clinic
2. Clinic staff search for patient using the search criteria SearchPatient(SearchCriteria)
3. The system returns a matching patient.
4. Clinic staff ask when the patient is available for an appointment. (checkAppointmentAvailability(patientID, professionalID, locationID, serviceID, requestedDateTime))
5. Booking an appointment checks that the doctor is available for the appointment and if there is need for a referral that it exists.
6. System returns availability is confirmed and referral requirements are completed. (availabilityConfirmed, referralRequirementsSatisfied)
7. Clinic Staff books the appointment with the details. (bookAppointment(patientID, professionalID, locationID, serviceID, requestedDateTime))
8. The system records the appointment and returns the confirmation. *(appointmentID, bookingConfirmation)* it sets the status to “Confirmed”, books the slot in the calendar and if needed it connects the referral.

Extensions

- 3a. Patient is not found
  - System returns there is no patient related to the search
  - Staff can then register a new patient
- 3b. Multiple Patients
  - Staff clicks correct patient from found list
- 5a. Requests a doctor that does not work at the clinic
  - System shows an error for the booking schedule
  - Staff then sees the location of the requested doctor and informs the patient of the new location
- 5b. Referral missing or invalid (Failure case 3)
- 5c. Schedule conflict or unavailable doctor

Special Requirements:

- Access: administrative staff only see registration and appointment data, no patient diagnosis information.
- Response time: under 5 seconds per operation.

Technology and Data Variations list:

- searchCriteria may be any:
    - patient ID;
    - insurance number (RAMQ or private);
    - date of birth;
    - phone number;
    - First or last name
- Patients can be covered by RAMQ or private insurance or can have no insurance. All patient booking processes are the same regardless of insurance status only invoicing is modified.
- Referrals may be done internally or externally.

Open Issues:

- Unsure how far away appointments can be booked.
- For cancellations, must it be at least 24 hrs in advance?

Use Case 1 (Failure):

Primary actor:
Clinic Staff

Stakeholders and Interest:

- Patient: Wants minimal wait time and wants a secure system to protect confidential info.
- Clinic Staff: Wants fast patient lookup, fast checks for doctor availability, schedule constraints and communication with receiving referrals between different clinics.
- Clinic Doctors: Accurate non-complicated software to input days off and prevent bookings on those days. Quick referral sending between clinics.

Preconditions:

- Clinic Staff or Clinic Doctors: are securely logged in with editing access to patient information or scheduling times.
- Clinic Doctors: want their schedules to be easily added to the system already.

Failure Guarantee (Postconditions):

- No appointment is created.
- The system schedule is not changed and the requested time is not blocked off.
- No referral is connected to an appointment.
- The rejected booking attempt is logged.

Main Scenario:

1. The patient requests an appointment at the clinic.
2. Clinic staff search for patients using the search criteria SearchPatient(SearchCriteria).
3. The system returns a matching patient.
4. Clinic staff ask when the patient is available for an appointment. (checkAppointmentAvailability(patientID, professionalID, locationID, serviceID, requestedDateTime))
5. The system checks if the doctor is available for the appointment and finds the doctor is not available at the requested time.
6. System returns the doctor is unavailable. (professionalUnavailable)
7. Clinic Staff still tries to book the appointment with the details. (bookAppointment(patientID, professionalID, locationID, serviceID, requestedDateTime))
8. The system rejects the booking and returns the reason. (bookingRejected, professionalUnavailable) No appointment is created.

Extensions:

- 3a. Patient is not found
    - System returns there is no patient related to the search.
    - Staff can then register a new patient.
- 3b. Multiple Patients
    - Staff clicks correct patient from found list.
- 5a. Requests a doctor that does not work at the clinic
    - System shows an error for the booking schedule.
    - Staff then sees the location of the requested doctor and informs the patient of the new location.

Special Requirements:

- Access: administrative staff only see registration and appointment data, no patient diagnosis information.
- Response time: under 5 seconds per operation.

Technology and Data Variations list:

- searchCriteria may be any:
    - patient ID;
    - insurance number (RAMQ or private);
    - date of birth;
    - phone number;
    - First or last name
- Patients can be covered by RAMQ or private insurance or can have no insurance. All patient booking processes are the same regardless of insurance status only invoicing is modified.
- Referrals may be done internally or externally.

Open Issues:

- Unsure how far away appointments can be booked.
- For cancellations, must it be at least 24 hrs in advance?

Use Case 2 (Success):

Primary actor:
Healthcare Professional

Stakeholders and Interest:

- Patient: Wants a quick consultation with minimal wait time. Wants their clinical information recorded accurately and kept confidential.
- Healthcare Professional: Wants quick access to the appointment and patient info. Wants simple, fast recording of observations, diagnoses, treatment recommendations and follow-ups.
- Clinic Staff: Wants to know when a consultation has started and finished, without seeing clinical information.
- Clinic: Wants complete, accurate clinical records and a log of who accessed or recorded patient information.

Preconditions:

- Healthcare Professional: is securely logged in with access to clinical information.
- The appointment is confirmed and assigned to the Healthcare Professional.
- Clinic Staff has confirmed the patient's arrival.

Success Guarantee (Postconditions):

- A consultation is created and linked to the appointment and the patient.
- The clinical information (observations, diagnoses, treatment recommendations, follow-up requirements) is saved in the patient's clinical record.
- The appointment status is updated to "Completed".
- The access to and recording of clinical information is logged.

Main Success Scenario:

1. Healthcare Professional opens the patient's appointment. (getAppointmentDetails(appointmentID))
2. The system returns the appointment details and patient information. (appointmentDetails, patientInformation)
3. Healthcare Professional starts the consultation. (startConsultation(appointmentID))
4. The system creates the consultation and returns its ID. (consultationID)
5. Healthcare Professional conducts the consultation with the patient.
6. Healthcare Professional records the clinical information. (recordClinicalInformation(consultationID, observations, diagnoses, treatmentRecommendations, followUpRequirements))
7. The system saves the clinical record, links it to the consultation and patient, sets the appointment to "Completed", and logs the activity. (clinicalRecordSaved)

Extensions:

- 1a. Appointment not found
    - The system returns no appointment with that ID or time.
    - Healthcare Professional re-enters the appointment ID or closes the view.
- 1b. Patient arrival not confirmed
    - The system returns that the patient has not arrived.
    - The Healthcare Professional asks Clinic Staff to confirm the arrival.
- 3a. Consultation already started for this appointment
    - The system returns the existing consultationID.
    - Healthcare Professional continues at step 5.
- 5a. Prescription needed
    - Healthcare Professional issues a prescription, then resumes at step 6.
- 5b. Follow-up appointment needed

    - Healthcare Professional records it in followUpRequirements. The follow-up is booked following the procedure from use Case 1.
- 6a. Clinical information missing
    - The system shows an error listing the missing fields.
    - Healthcare Professional fills them in and resumes at step 6.

Special Requirements:

- Access: only authorized Healthcare Professionals can view and record clinical information. Clinic Staff see no patient diagnosis information.
- Response time: under 5 seconds per operation.
- Every view and save of clinical information is logged.

Technology and Data Variations list:

- Clinical information include:
    - observations
    - diagnoses
    - treatment recommendations
    - follow-up requirements
- Electronic prescriptions to pharmacies and

Open Issues:

- Can a clinical record be edited after it is saved, and by whom?
- Which informational fields are required versus optional?
- What happens to the appointment if the patient never arrives (no-show)?

Use Case 2 (Failure):

Primary actor:
Administrative Staff

Stakeholders and Interest:

- Patient: Wants a quick consultation with minimal wait time. Wants their clinical information recorded accurately and kept confidential.
- Healthcare Professional: Wants quick access to the appointment and patient info. Wants simple, fast recording of observations, diagnoses, treatment recommendations and follow-ups.
- Clinic Staff: Wants to know when a consultation has started and finished, without seeing clinical information.
- Clinic: Wants complete, accurate clinical records and a log of who accessed or recorded patient information.

Preconditions:

- Administrative Staff: is securely logged in without access to clinical information.
- The appointment is confirmed and assigned to the Healthcare Professional.
- Clinic Staff has confirmed the patient's arrival.

Failure Guarantee (Postconditions):

- No clinical record is created or modified.
- The appointment status is not changed.
- The denied attempt to record clinical information is logged.

Main Scenario:

1. Administrative Staff opens the patient's appointment. (getAppointmentDetails(appointmentID))
2. The system returns the appointment details only, with no clinical information. (appointmentDetails)
3. Administrative Staff tries to record clinical information. (recordClinicalInformation(consultationID, observations, diagnoses, treatmentRecommendations, followUpRequirements))
4. The system checks permissions, denies access, and logs the attempt. (accessDenied, insufficientPermissions) No clinical record is created or modified.

Extensions:

- 1a. Appointment not found
    - The system returns no appointment with that ID or time.
    - Administrative Staff re-enters the appointment ID or closes the view.

Special Requirements:

- Access: only authorized Healthcare Professionals can view and record clinical information. Clinic Staff see no patient diagnosis information.
- Response time: under 5 seconds per operation.
- Every view and save of clinical information is logged.

Technology and Data Variations list:

- Clinical information include:
    - observations
    - diagnoses
    - treatment recommendations
    - follow-up requirements
- Electronic prescriptions to pharmacies

Open Issues:

- Who has the access to edit a clinical record after it is saved?
- Which informational fields are required versus optional?
- What happens to the appointment if the patient never arrives?

Use Case 3 (Success):

Primary actor:
Healthcare Professional

Stakeholders and Interest:

- Patient: Wants to be referred quickly to the right specialist. Wants their clinical reason kept confidential.
- Healthcare Professional: Wants fast patient lookup and quick referral sending.
- Receiving Specialist: Wants clear referrals with the clinical reason and priority.
- Clinic Staff: Wants valid referrals in the system so appointments that require one can be booked.

- Clinic: Wants accurate referral records and a log of who issued them.

Preconditions:

- Healthcare Professional: is securely logged in with access to clinical information.
- The patient is registered in the system.

Success Guarantee (Postconditions):

- A referral is created and linked to the patient and the referring professional.
- The requested specialty and/or professional, clinical reason, and priority are recorded.
- The referral status is set to "Pending" (not yet linked to an appointment).
- The referral is logged.

Main Success Scenario:

1. Healthcare Professional searches for the patient. (searchPatient(searchCriteria))
2. The system returns the matching patient information. (patientInformation)
3. Healthcare Professional issues the referral. (issueReferral(patientID, referringProfessionalID, requestedSpecialtyID, requestedProfessionalID, clinicalReason, priority))
4. The system creates the referral, sets its status, logs it, and returns the confirmation. (referralID, referralStatus, referralConfirmation)

Extensions:

- 2a. Patient is not found
    - The system returns no patient related to the search.
    - Clinic Staff registers the patient, then Healthcare Professional resumes at step 1.
- 2b. Multiple Patients
    - Healthcare Professional selects the correct patient from the list.
- 3a. No specific professional requested
    - Healthcare Professional leaves requestedProfessionalID empty and only the specialty is recorded.
- 3b. Required referral information missing
    - The system shows an error listing the missing fields.
    - Healthcare Professional fills them in and resumes at step 3.

Special Requirements:

- Access: only authorized Healthcare Professionals can issue referrals.
- Response time: under 5 seconds per operation.
- Every referral issued is logged.

Technology and Data Variations list:

- searchCriteria may be any:
    - patient ID
    - insurance number (RAMQ or private)
    - date of birth
    - phone number
    - First or last name
- Referring professionals may be internal or external to the clinic.
- Priority levels.

Open Issues:

- How long is a referral valid?
- How many appointments can one referral be used for?
- How are external professionals' referrals entered into the system?

Use Case 3 (Failure):

Primary actor:
Healthcare Professional

Stakeholders and Interest:

- Patient: Wants to be referred quickly to the right specialist. Wants their clinical reason kept confidential.
- Healthcare Professional: Wants fast patient lookup and quick referral sending.
- Receiving Specialist: Wants clear referrals with the reason and priority.
- Clinic Staff: Wants valid referrals in the system so appointments that require one can be booked.
- Clinic: Wants accurate referral records and a log of who issued them.

Preconditions:

- Healthcare Professional: is securely logged in with access to clinical information.
- The patient is registered in the system.

Failure Guarantee (Postconditions):

- No referral is created.
- Nothing is linked to the patient or the referring professional.
- The rejected referral attempt is logged.

Main Scenario:

1. Healthcare Professional searches for the patient. (searchPatient(searchCriteria))
2. The system returns the matching patient information. (patientInformation)
3. Healthcare Professional issues the referral without a requested specialty or a requested professional. (issueReferral(patientID, referringProfessionalID, requestedSpecialtyID, requestedProfessionalID, clinicalReason, priority))
4. The system rejects the referral, logs the attempt, and returns the reason. (referralRejected, missingRequiredInformation) No referral is created.

Extensions:

- 2a. Patient is not found
    - The system returns no patient related to the search.
    - Clinic Staff registers the patient, then Healthcare Professional resumes at step 1.
- 2b. Multiple Patients
    - Healthcare Professional selects the correct patient from the list.
- 4a. Healthcare Professional corrects the referral
    - Healthcare Professional requests the patient to get a referral or for the healthcare specialist to resubmit the referral.
    - The referral continues as Use Case 3 (Success).

Special Requirements:

- Access: only authorized Healthcare Professionals can issue referrals.
- Response time: under 5 seconds per operation.
- Every referral issued is logged.

Technology and Data Variations list:

- searchCriteria may be any:
    - patient ID
    - insurance number (RAMQ or private)
    - date of birth
    - phone number
    - First or last name
- Referring professionals may be internal or external to the clinic.
- Priority levels.

Open Issues:

- How long is a referral valid?
- How many appointments can one referral be used for?
- How are external professionals' referrals entered into the system?

5. System operations and operation contracts

System Operations

UC1: Book Appointment

- searchPatient(searchCriteria)
- checkAppointmentAvailability(patientID, professionalID, locationID, serviceID, requestedDateTime)
- bookAppointment(patientID, professionalID, locationID, serviceID, requestedDateTime)

UC2: Conduct Consultation

- getAppointmentDetails(appointmentID)
- startConsultation(appointmentID)
- recordClinicalInformation(consultationID, observations, diagnoses, treatmentRecommendations, followUpRequirements)

UC3: Issue Referral

- searchPatient(searchCriteria)
- issueReferral(patientID, referringProfessionalID, requestedSpecialtyID, requestedProfessionalID, clinicalReason, priority)

System Interface

System
• searchPatient(searchCriteria) • checkAppointmentAvailability(patientID, professionalID, locationID, serviceID, requestedDateTime) bookAppointment(patientID, professionalID, locationID, serviceID, requestedDateTime) getAppointmentDetails(appointmentID) • startConsultation(appointmentID) • recordClinicalInformation(consultationID, observations, diagnoses, treatmentRecommendations, followUpRequirements) • issueReferral(patientID, referringProfessionalID, requestedSpecialtyID, requestedProfessionalID, clinicalReason, priority)

Operation Contracts

Contract CO1: searchPatient

- Operation:
  - searchPatient(searchCriteria: SearchCriteria)
- Cross References:
  - Use Cases: UC1 Book Appointment, UC3 Issue Referral
- Preconditions:
  - The user is logged in.
- Postconditions:
  - None.

Contract CO2: checkAppointmentAvailability

- Operation:
  - checkAppointmentAvailability(patientID: PatientID, professionalID: ProfessionalID, locationID: LocationID, serviceID: ServiceID, requestedDateTime: DateTime)
- Cross References:
  - Use Case: UC1 Book Appointment
- Preconditions:
  - Patient exists.
  - HealthcareProfessional exists.
  - ClinicLocation exists.
  - Service exists.
- Postconditions: None (query operation).

Contract CO3: bookAppointment

- Operation:
  - bookAppointment(patientID: PatientID, professionalID: ProfessionalID, locationID: LocationID, serviceID: ServiceID, requestedDateTime: DateTime)
- Cross References:

    - Use Case: UC1 Book Appointment
- Preconditions:
    - Patient exists.
    - HealthcareProfessional exists and practises at the ClinicLocation.
    - HealthcareProfessional is available at requestedDateTime.
    - A valid Referral exists if the Service requires one.
- Postconditions:
    - An Appointment instance (*apt*) was created. (instance creation)
    - *apt*.dateTime was set to requestedDateTime. (attribute modification)
    - *apt*.status was set to "Confirmed". (attribute modification)
    - *apt* was associated with the Patient, based on patientID match. (association formed)
    - *apt* was associated with the HealthcareProfessional, based on professionalID match. (association formed)
    - *apt* was associated with the ClinicLocation, based on locationID match. (association formed)
    - *apt* was associated with the Service, based on serviceID match. (association formed)
    - *apt* was associated with the Referral, if the Service requires one. (association formed)
    - The time slot in the HealthcareProfessional's Schedule was set to booked. (attribute modification)

Contract CO4: getAppointmentDetails

- Operation:
    - getAppointmentDetails(appointmentID: AppointmentID)
- Cross References:
    - Use Case: UC2 Conduct Consultation
- Preconditions:
    - Appointment exists.
- Postconditions:
    - None (query operation).

Contract CO5: startConsultation

- Operation:
    - startConsultation(appointmentID: AppointmentID)
- Cross References:
    - Use Case: UC2 Conduct Consultation
- Preconditions:
    - Appointment exists with status "Confirmed".
    - Appointment is assigned to the HealthcareProfessional.
    - Patient arrival has been confirmed.
- Postconditions:
    - A Consultation instance *consult* was created. (instance creation)
    - *consult*.startTime was set to the current date and time. (attribute modification)
    - *consult* was associated with the Appointment, based on appointmentID match. (association formed)
    - *consult* was associated with the HealthcareProfessional. (association formed)
    - Appointment.status was set to "In Progress". (attribute modification)

Contract CO6: recordClinicalInformation

- Operation:
  - recordClinicalInformation(consultationID: ConsultationID, observations: String, diagnoses: String, treatmentRecommendations: String, followUpRequirements: String)
- Cross References:
  - Use Case: UC2 Conduct Consultation
- Preconditions:
  - Consultation exists.
  - The user is an authorized HealthcareProfessional.
- Postconditions:
  - A ClinicalRecord instance *cr* was created. (instance creation)
  - *cr*.observations was set to observations. (attribute modification)
  - *cr*.diagnoses was set to diagnoses. (attribute modification)
  - *cr*.treatmentRecommendations was set to treatmentRecommendations. (attribute modification)
  - *cr*.followUpRequirements was set to followUpRequirements. (attribute modification)
  - *cr* was associated with the Consultation, based on consultationID match. (association formed)
  - *cr* was associated with the Patient. (association formed)
  - Appointment.status was set to "Completed". (attribute modification)

Contract CO7: issueReferral

- Operation: issueReferral(patientID: PatientID, referringProfessionalID: ProfessionalID, requestedSpecialtyID: SpecialtyID, requestedProfessionalID: ProfessionalID, clinicalReason: String, priority: Priority)
- Cross References: Use Case: UC3 Issue Referral
- Preconditions:
  - Patient exists.
  - Referring HealthcareProfessional exists.
  - A requested Specialty or requested HealthcareProfessional is specified.
- Postconditions:
  - A instance *referral* was created. (instance creation)
  - *referral*.clinicalReason was set to clinicalReason. (attribute modification)
  - *referral*.priority was set to priority. (attribute modification)
  - *referral*.status was set to "Pending". (attribute modification)
  - *referral* was associated with the Patient, based on patientID match. (association formed)
  - *referral* was associated with the referring HealthcareProfessional, based on referringProfessionalID match. (association formed)
  - *referral* was associated with the requested Specialty, based on requestedSpecialtyID match, if specified. (association formed)
  - *referral* was associated with the requested HealthcareProfessional, based on requestedProfessionalID match, if specified. (association formed)